

10th International Conference of Information and Communication Technology (ICICT-2020)

Performance analysis of big data transmission in wearable system based on special textile clothing

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Abstract

Special textile clothing covers a wide range. It includes helmet, vest, coat, trousers and boots. In the home environment, wearable devices can monitor the physiological parameters of the elderly or children according to the requirements. These physiological parameters include EEG, heart rate, blood pressure, blood glucose, height, weight, walking speed and standing posture. The acquisition and remote transmission of these parameters are very important for the monitored and information center. Therefore, it is necessary to analyze and simulate the transmission performance of data. This work mainly introduces the principle of performance analysis, simulation model and numerical calculation. Simulation results show that the proposed scheme can still improve the quality of service under the same network conditions.

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Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Blood pressure, standing posture, remote transmission, acquisition

1. Introduction

In case of emergency, the data collection type of wearable device should be different. Under normal conditions, height and weight parameters will occupy most of the data network resources. In the case of emergency, such as the rapid heartbeat and insufficient blood supply, the transmission of parameters must be distinguished and selected. Therefore, priority data transmission rules should be considered and applied. Since most of the data generated are random signals, the analysis of such signals becomes particularly important. The analysis parameters include mean

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value, variance, autocorrelation function, cross-correlation function and power spectral density. Because of the independence of the data quantity between the former time and the latter time, the conclusion of the analysis of the definite signal is no longer suitable for this situation.

We assume that data is collected by wireless sensors. The data type collected by each node is determined by the function of the device. Then it converges to the mobile terminal. The network of the mobile terminal can be 4G network or 5g network, which depends on the data transmission environment. A typical wearable sensor network is shown in Figure 1.

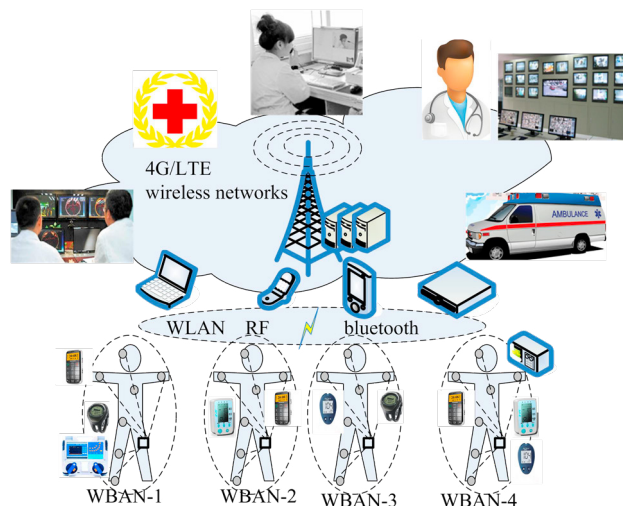


Fig. 1. Wearable sensor network architecture.

2. Performance definition

The importance of human body parameters is divided into different categories. Priority allocation of network resources for emergency data parameters. The data flow direction is shown in Fig. 2.

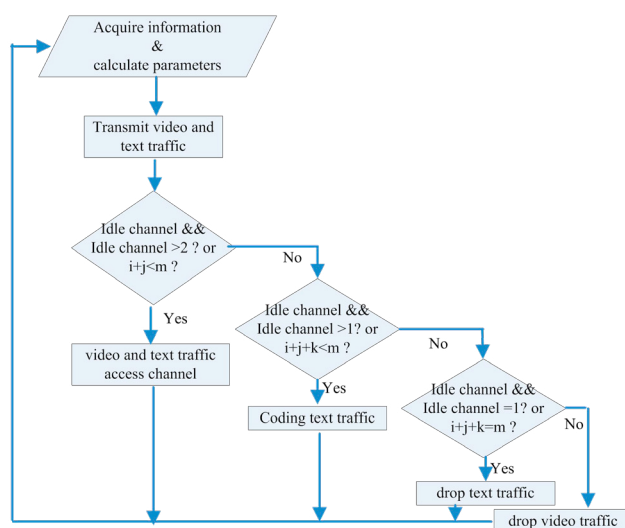


Fig. 2. Logic diagram of data flow processing

2.1. Performance analysis

Data streams can be defined as:

$$\begin{aligned}
 & Por(Q(u1, u2) - e(u2 - u1) > a) \\
 &= \sum_{1 \text{ and } 2} Por(\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} + \{A_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\
 &\leq \inf_{z < x} (Por\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\
 &\quad + \{Q_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\
 &\leq \inf_{z < x} (A_1(K) + A_2(a - K)) \\
 &= \sum_{< 2} (A_1 \otimes A_2)(a)
 \end{aligned} \tag{1}$$

and

$$\begin{aligned}
 & Por(Q(u1, u2) - e(u2 - u1) - e(u3 - u2) - e(u4 - u3) > a) \\
 &= \sum_{> 2} Por(\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} + \{A_2(u1, u2) - \alpha_2(u2 - u1)\} > a) \\
 &\leq \inf_{z < x} (Por\{Q_1(u1, u2) - \alpha_1(u2 - u1)\} > K \\
 &\quad + \{Q_2(u1, u2) - \alpha_2(u2 - u1)\} > a - K) \\
 &\leq \inf_{z < x} (A_1(K) + A_2(a - K)) \\
 &= \sum_{> 2} (A_1 \otimes A_2)(a) \\
 &\leq (A_1 \otimes \dots \otimes \dots \otimes A_N)(a)
 \end{aligned} \tag{2}$$

2.2. Steady state

The system state transitions in S can be depicted in Fig. 3.

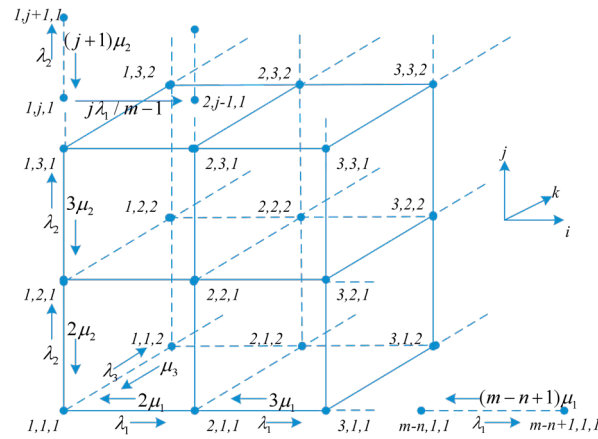


Fig. 3. Schematic diagram of data migration in 3D state

3. Data service Evaluation

The procedure of model construction is shown in Fig.4.

```

1: Initialize ;
2: if
3:   else if
4:     Data generation;
5:     else if
6:       vector of density values
7:     end if
8:   end if
9: end if
10: for (0: normal kernel function)
11:   for (1: spaced points)
12:     for (2: window parameter)
13:       specifies the vector;
14:
15: name/value;
16: for (0: censoring times)
17:   function ;
18:   end if
19: end if
20: cumulative distribution
21: estimation function
22: density function
23: n-by-h matrix;
24: model parameters
25: single row;
26: fixed effects
27: n-by-1 vector;

```

Fig. 4. Transition probability calculation

4. Analysis of transition probability

Comparison of termination probabilities is shown in Fig.5.

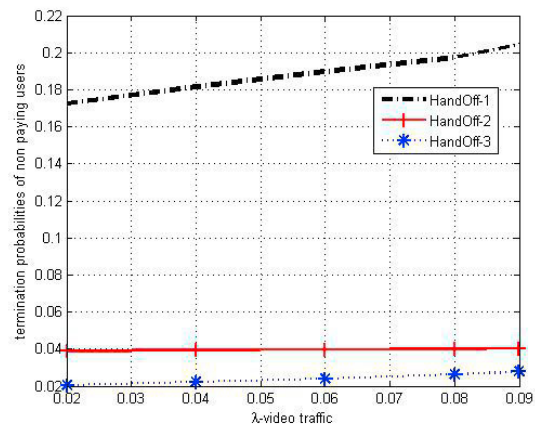


Fig. 5. Termination probabilities

Blocking probabilities can be represented in Figure 6.

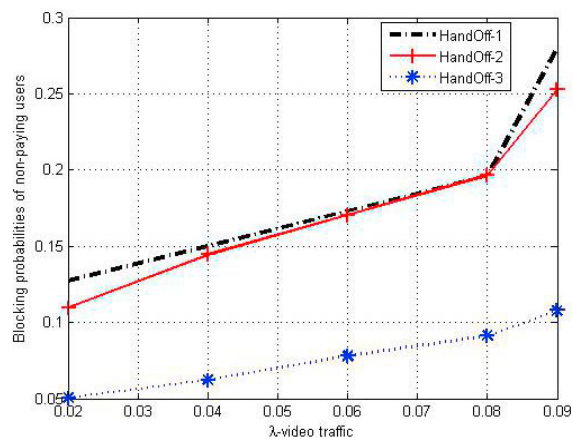


Fig. 6. Blocking probabilities.

System throughput is shown as in Figure 7.

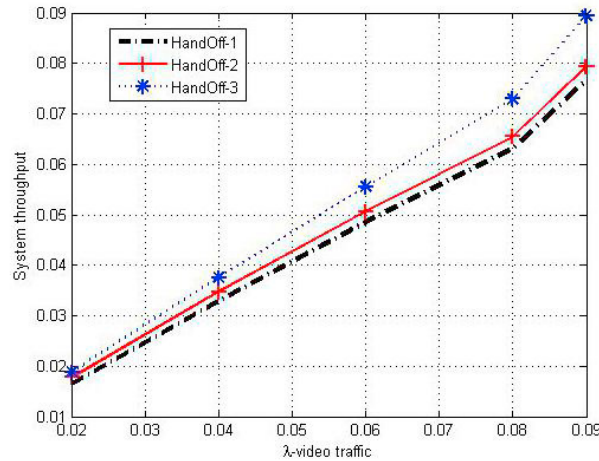


Fig. 7. System throughput

Spectrum resource utility can be represented in Figure 8.

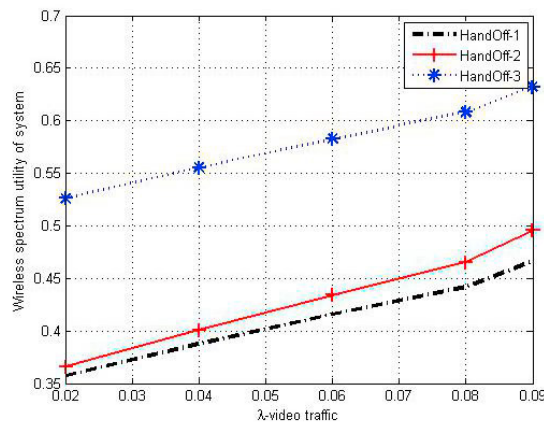


Fig. 8. Spectrum resource utility

Mechanism HandOff-1 in the graph is always kept in the interval of higher values. The average probability of mechanism HandOff-2 and mechanism HandOff-3 fluctuated around 0.03. Therefore, it can be found that the proposed mechanism can significantly improve the video transmission capacity of the system for outage probability. It can be seen from Fig. 6 that the blocking probability of mechanism HandOff-3 is significantly better than that of the other two mechanisms. In this scenario, mechanism HandOff-1 and mechanism HandOff-2 always maintain a high blocking probability. When the video data rate is close to 0.08, the blocking probability will increase greatly. It means that the transmission of video data faces greater uncertainty. In contrast, the mechanism proposed in this paper can effectively improve the occurrence of system blocking probability, and maintain a very flat curve distribution throughout the transmission process. When the arrival rate of video data changes from 0.02 to 1, the blocking probability of the proposed mechanism is limited at about 10%. This value will make the transmission of video data more smooth and stable.

Figure 7 shows the trend of the overall throughput for the system under different mechanisms. With the increase of input data, the system throughput corresponding to the three forwarding mechanisms is increasing, and the

improvement of mechanism HandOff-3 is the most obvious. According to the change trend of the overall throughput of the system, when the data input rate is 0.02, the throughput of mechanism HandOff-3 is 2% higher than that of mechanism 1. When the data input rate is increased to 0.04, the throughput of mechanism HandOff-3 is 8.1% higher than that of mechanism HandOff-1, and 6.2% higher than that of mechanism HandOff-2. When the data input rate increases to 0.08, the throughput of mechanism HandOff-3 is improved more obviously. At this time, the throughput of mechanism HandOff-3 is 10.3% higher than that of mechanism HandOff-1, and the 8.5% higher than that of mechanism HandOff-2. When the data input rate is close to 1, the throughput improvement of mechanism HandOff-3 is about 15.3% higher than that of mechanism HandOff-1, and the 10.2% higher than that of mechanism HandOff-2. Through the above comparison, we can see that the mechanism proposed in this paper can adapt to the scenario of large data flow and avoid the loss of data packets.

Figure 8 shows a comparison of the system spectrum resource utilization. From the overall change trend, the system spectrum utilization under the three mechanisms increases with the data input stream, which is also in line with the actual operation of the system. The mechanism proposed in this paper can better reflect the utilization ratio of wireless spectrum resources. When the data input flow rate is 0.02, the spectrum efficiency of mechanism HandOff-3 is higher than 20% of mechanism HandOff-1. When the data input flow rate is close to 1, the spectrum efficiency of mechanism 3 is higher than that of mechanism 1 by 23.4%.

5. Conclusions

This paper studies a variety of performance parameters of wearable network. The throughput and spectrum utilization are compared in detail. It can be seen from the conclusion that the wireless sensor network meets the design expectation for different priority data. Numerical simulation results show that the network utilization is close to the optimal value under different data types. Moreover, when the data conflict, the data with high priority can occupy the wireless channel resources quickly and stably.

Acknowledgement

Acknowledgement Science and technology department of Hubei province (No. 2019ADC033), Hubei provincial department of education (No. 2019GB033), Hubei Key Laboratory of Digital Textile Equipment(Wuhan Textile University). Graduate Department of Wuhan Textile University (No. 201901008, No. 2020JY053, 2020JY075).

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